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ADAS_Radar2.py
1 import cv2
2 import numpy as np
3 import RPi.GPIO as GPIO
4 import time
5
6 # X-band radar sensing output connected to GPIO 27 of Rpi
7 HB100_INPUT_PIN = 27
8
9 # configure GPIO pins for input into GPIO 27
10 GPIO.setmode(GPIO.BCM)
11 GPIO.setup(HB100_INPUT_PIN, GPIO.IN)
12
13 # sample global vars for adjustment of sensitivity/measurements
14 # - A greater max-pulse-count will count more pulse inputs, leading
15 #   to more averaged doppler frequency - it will also take longer
16 # - A higher motion-sensitivity will reduce false motion readings
17 MAX_PULSE_COUNT = 10
18 MOTION_SENSITIVITY = 10
19
20 cap = cv2.VideoCapture(0) # initialize the camera
21
22 safe_distance_color = (0, 255, 0) # Green color
23 safe_distance_thickness = 2
24 safe_distance_position = 280 # Adjust the vertical position as needed
25
26
27 def count_frequency(GPIO_pin, max_pulse_count=10, ms_timeout=50):
28     """ Monitors the desired GPIO input pin and measures the frequency
29         of an incoming signal. For this example it measures the output of
30         a HB100 X-Band Radar Doppler sensor, where the frequency represents
31         the measured Doppler frequency due to detected motion.
32     """
33     start_time = time.time()
34     pulse_count = 0
35
36     # count time it takes for 10 pulses
37     for count in range(max_pulse_count):
38         # wait for falling pulse edge - or timeout after 50 ms
39         edge_detected = GPIO.wait_for_edge(GPIO_pin, GPIO.FALLING, timeout=ms_timeout)
40
41         # if pulse detected - iterate count
42         if edge_detected is not None:
43             pulse_count += 1
44             start_time = time.time()
45
46     return pulse_count

```

Fig. 7: Code snippet of ADAS system and radar

The doppler effect occurs because when an object emitting waves (of sound or light) is in motion relative to an observer, the waves emitted by the object get compressed in front of the object and are stretched behind

it. This results in a change in the observed frequency or wavelength of the waves perceived by the observer. There are two main types of doppler effects for sound:

Moving source and stationary observer.

If the source of the sound is moving toward the observer, the sound waves are

compressed, and the observed frequency is higher (producing a higher-pitched sound). If the source is moving away from the observer, the sound waves are stretched, and the observed frequency is lower (producing a lower-pitched sound).

Stationary source and moving observer. If the observer is moving toward the stationary sound source, the same effect occurs, and the observed frequency is higher. If the observer is moving away from the stationary source, the observed frequency is lower.

These radar systems work on the same fundamental principles as other doppler radar applications but are adapted for use in vehicles. Here is how the doppler radar works in a vehicle:

Transmitting radar signals.

The vehicle's radar system emits radio frequency (RF) signals in a specific direction, typically in the microwave frequency range. These signals are transmitted using radar antennas or sensors located on the vehicle's front grille or bumpers.

Reflection and doppler shift.

When the radar signals encounter objects in the vehicle's path, such as other vehicles, pedestrians, or obstacles, some of the RF energy is reflected back towards the radar antenna.

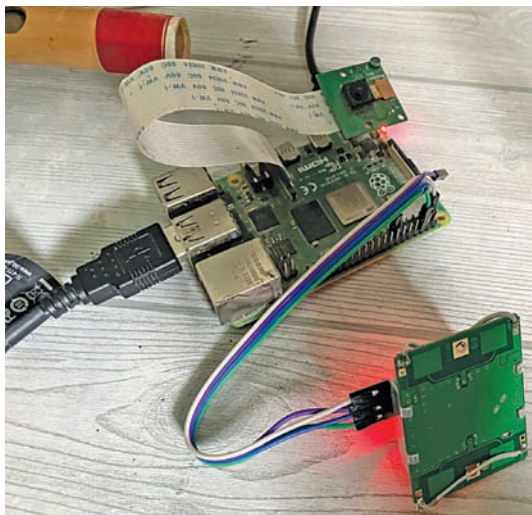


Fig. 8: The author's setup during testing



Fig. 9: Radar and camera attached on bumper of car